



Topic:

API Framework — AI-Generated REST Test Automation

Scope:

The API Framework is an AI-powered REST automation suite that converts a JSON endpoint catalog (the single source of truth) into a fully executable RestAssured + TestNG Java suite. A Node.js generator on port 3001 sends the catalog to Anthropic Claude, which returns scenarios + test cases covering 5 mandatory categories (positive, auth, validation, non-existent, pagination) and a single Java class with one `@Test` per case. Maven runs the suite, a TestNG listener captures per-test status into `test_results.json`, ExcelJS surgically updates a 4-sheet Excel report (Scenarios, Test Cases, Defect Report, RTM), failed tests are narrowly re-run via `-Dtest=Class#TC_001*+...`, and an Allure HTML report is generated + opened — all from one input file.

Key Features of the API Framework:

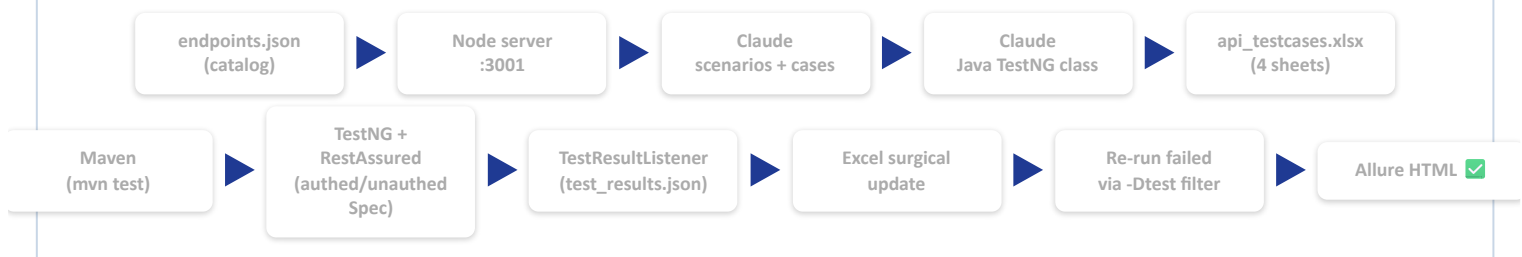
- **Two-Pass AI Generation:** Claude (Sonnet 4.6) first returns `{scenarios, testCases}` JSON; a second call produces a single RestAssured + TestNG Java class with one `@Test` per case — no hand-written tests.
- **5 Mandatory Coverage Categories:** Positive (200), Auth-negative (401/403), Validation (422), Non-existent (404), Pagination (limit/offset) — enforced on every run; scales linearly with catalog size (~60–80 tests for ~50 endpoints).
- **Session-Cookie Auth Done Once:** `BaseApiTest` performs CSRF scrape + form login; exposes `authenticatedSpec` (with cookie) and `unauthenticatedSpec` RequestSpecifications reused across every test.
- **4-Sheet Excel Report:** Test Scenarios, Test Cases, Defect Report (auto-populated for failures with Sev/Pri/Status), and RTM linking Requirement → Scenario → Test Case → Defect.
- **Surgical Excel Updates:** Only rows present in `test_results.json` get touched; Defect IDs are SET for fails and CLEARED for tests that pass on retry; other rows untouched.
- **Triage + Narrow Retry:** Failed tests are re-run via Maven Surefire filter `-Dtest=Class#TC_001*+TC_005*+...`; up to `MAX_RETRY_ATTEMPTS = 2`; no LLM patch (API failures are data/auth/transient — re-execution suffices).
- **Allure Integration:** Per-test request/response capture via `ResponseCapture.java`, full timeline, test history; HTML report opens automatically.
- **Single Source of Truth:** Add/remove/edit endpoints in `endpoints.json` only — the test suite, Excel, Defect Report and RTM regenerate from it.
- **Two Run Modes:** Stand-alone `run-tests.bat` (fully serial 7-step pipeline) or Master Orchestrator integration via `node index.js --framework=api`.

Why Use the API Framework?

Hand-writing REST tests is slow and goes stale the moment an endpoint is added. The API Framework keeps coverage in lock-step with the catalog: drop an endpoint into `endpoints.json` and the test suite, Excel report, Defect Report and RTM update on the next run. The five enforced categories guarantee no endpoint ships untested for auth, validation, missing-resource or pagination behaviour. Because retries are scoped to failed methods only (Maven Surefire `-Dtest=` filter), large suites stay fast — full-suite cost is paid once, retries are cheap. Surgical Excel updates mean passing rows stay green across retries and only the changed defect IDs move. Integration with the Master Orchestrator lets the same catalog drive the UI suite and Jira manual cases in one session.

API Framework Architecture

Two pipelines run end-to-end: **Generation** (catalog → Claude → Java + Excel) and **Execution + Triage** (Maven → TestNG listener → surgical Excel → narrow retry → Allure).



Integration with Master Orchestrator, Tooling & CI/CD

Plugs into the Master Orchestrator via `apiAdapter.js` (`node index.js --framework=api`) — drop `endpoints.json` into `master-orchestrator/inputs/api/` and the orchestrator runs prep → generate → execute → triage → archive → consolidated Zephyr push. Stand-alone runs use `run-tests.bat`'s serial 7-step pipeline. Built on **Node.js + Express + ExcelJS + dotenv** for the generator and **Java + Maven + TestNG + RestAssured + Allure** for the runner; Anthropic Claude (Sonnet 4.6) handles both generation passes. Per-test request/response is auto-attached to Allure for debugging. The `test_results.json` output feeds the orchestrator's surgical Zephyr update at session end and feeds the orchestrator's healing-bridge summary. Maven + Allure integrate cleanly with Jenkins, GitHub Actions, GitLab CI and Azure DevOps.

References: docs.anthropic.com · rest-assured.io · testng.org · docs.qameta.io/allure · maven.apache.org

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